Total No. of Questions : 8]

SEAT No. :

PE2168

[Total No. of Pages : 3

[6584]-67
B.E. (Civil Engg.)
FOUNDATION ENGINEERING
(2019 Pattern) (Semester - VII) (401001)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn whenever necessary.***
3) ***Figures to the right indicate full Marks.***
4) ***Assume suitable data, if necessary and mention it clearly.***
5) ***Use of non-programmable calculator is allowed.***

Q1) a) Explain the term: **[5]**

i) Coeff. of Compressibility.

ii) Compression Index.

iii) Coeff. of volume Compressibility.

iv) Degree of Consolidation

v) Coeff. of Consolidation.

b) A soil stratum is 10 m thick with pervious stratum on top and bottom. Determine the time required for 50% consolidation. Given that coefficient of permeability = 10–7 cm/s, coefficient of compression = 0.0003 cm^2/gm, void ratio = 2 and time factor = 0.197. **[6]**

c) Explain with neat sketch laboratory consolidation test. **[6]**

OR

Q2) a) What is contact pressure? Draw contact pressure distribution of a rigid footing on sandy and clayey soil strata. **[6]**

b) In a consolidation test void ratio decreased from 0.75 to 0.70, when the load was changed from 50kN/m^2. Compute compression index of volume change. **[5]**

c) Discuss with sketch Square root of time fitting method for determination of coefficient of consolidation. **[6]**

P.T.O.

Q3) a) What are the conditions where a pile foundation is more suitable than a shallow foundation? **[6]**

b) A group of 9 piles, 10m long is used as a foundation for bridge pier. The piles used are 30 cm diameter with centre to centre spacing of 0.9m. the subsoil consists of clay with unconfined compressive strength of 1.5kg/cm^2. Determine the efficiency neglecting the bearing action. Assume adhesion factor = 0.9. **[6]**

c) Explain in brief efficiency of pile group. **[5]**

OR

Q4) a) How do you classify piles according to different criteria? **[6]**

b) A group of 16 piles of 50 cm diameter is arranged with a centre to centre spacing of 1.0m. The piles are 9m long and are embedded in soft clay with cohesion 30kN/m^2. Bearing resistance may be neglected for the piles (Adhesion factor 0.6). Determine the ultimate load capacity of the pile group. **[6]**

c) Define Negative Skin Friction. How it is determined. Also explain how it is prevented. **[5]**

Q5) a) Discuss the procedure for proportioning of footing for equal settlement. **[6]**

b) Discuss tilts and shifts: precautionary and remedial measures of well Foundation. **[6]**

c) Explain design steps of raft foundation by elastic (flexible) method. **[6]**

OR

Q6) a) What is meant by shallow foundations? What are the principles of the design of footing? **[6]**

b) What is Caisson? How Caissons are classified based on methods of construction? **[6]**

c) Sketch and describe the various components of the well foundation, indicating functions of each component. **[6]**

Q7) a) Explain the effect of swelling and shrinkage of expansive soils on building constructed. Also, enlist the precautions to be taken. Illustrate with sketches. **[6]**

b) Discuss design principle of undreamed pile. **[6]**

c) List out the various techniques of soil improvement. Explain any one. **[6]**

OR

Q8) a) Discuss preloading with prefabricated vertical drains/sand drains with a neat sketch. **[6]**

b) Explain any four engineering problem associated with black cotton soil. **[6]**

c) Enlist types of cofferdams. Explain any one type with neat sketch. **[6]**

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